

Jun Ma

PhD Candidate in Computational Mathematics
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Research Profile

Doctoral researcher working at the interface of topology optimization, PDE analysis, and AI-assisted scientific computing. Current research focuses on nonlinear Kapitza interface laws, bifurcation and non-uniqueness in heat conduction problems, and physics-informed neural networks for discovering multiple solutions of nonlinear PDEs.

Education

2024–present	PhD in Computational Mathematics Université de Technologie de Compiègne, France LMAC Laboratory. Supervisors: Prof. Faker Ben Belgacem and Prof. Faten Jelassi.
2021–2024	MSc in Computational Mathematics Northwestern Polytechnical University, China School of Mathematics and Statistics. Supervisors: Academician Weihong Zhang, Prof. Manyu Xiao, and Prof. Piotr Breitkopf.
2017–2021	BSc in Mathematics and Applied Mathematics Northwestern Polytechnical University, China School of Mathematics and Statistics.

Research Interests

- Topology and shape optimization; large-scale structural dynamics.
- PDE analysis, bifurcation, nonlinear interface laws, and Kapitza contact resistance.
- Reduced-order modeling, primal-dual POD, and adaptive model updating.
- Physics-informed neural networks and AI-assisted scientific computing.
- Thermal energy systems, phase change materials, and computational mechanics.

Selected Publications

1. Bejaoui, E., Ben Belgacem, F., Jelassi, F., and Ma, J. (2026). “Semilinear Elliptic Problems with NonLinear Contact Resistance in Composite Domains.” *HAL preprint*.

2. Xiao, M., Ma, J., Gao, X., et al. (2024). “Primal-dual on-the-fly reduced-order modeling for large-scale transient dynamic topology optimization.” *Computer Methods in Applied Mechanics and Engineering*, 428: 117099.

3. Bejaoui, E., Ben Belgacem, F., Jelassi, F., et al. (2026). “Steady conduction problems with non-linear Kapitza contact resistance: Existence and non-uniqueness.” *Mathematics and Computers in Simulation*.

4. Xiao, M., Ma, J., Lu, D., et al. (2022). “Stress-constrained topology optimization using approximate reanalysis with on-the-fly reduced order modeling.” *Advanced Modeling and Simulation in Engineering Sciences*, 9(1): 1–30.

5. Ma, R., Ma, J., and Xiao, M. (2022). “Several Computational Methods for Geodesic Equations on Surfaces.” *College Mathematics*, 38(4).

Current Doctoral Research

Kapitza laws	Nonlinear contact resistance and bifurcation analysis Existence, non-uniqueness, bifurcation, and hysteresis phenomena in steady heat conduction across composite-domain interfaces with temperature-dependent contact resistance.
Numerics	Galerkin and finite element approximation Finite element analysis for linear and nonlinear Kapitza interface models, with FreeFEM implementations and ongoing work on discrete maximum-principle and L^∞ estimates.
AI for PDEs	PINN-based multi-solution discovery Probabilistic framework where PINNs serve as randomized seed generators, followed by Newton/FEM refinement and certification of multiple nonlinear PDE solutions.

Doctoral Progress by Year

Year 1

Foundations in TES modeling, Kapitza laws, and topology optimization

Derived and numerically verified steady and transient heat-conduction models; implemented one- and two-dimensional FreeFEM++ simulations; incorporated radiative heat transfer through Stefan-Boltzmann-type nonlinear terms; studied nonlinear Kapitza contact resistance in double-layer domains; demonstrated non-uniqueness in scalar models; used Picard iteration and bifurcation analysis to identify transcritical, pitchfork, flip, and saddle-node bifurcations; studied topological derivatives via Dirichlet-to-Neumann operators and implemented a two-dimensional transient heat-conduction topology optimization prototype.

Year 2

Kapitza FEM, continuation, topological derivatives, and XPINN methods

Developed FreeFEM and Matlab tools for linear and nonlinear Kapitza interface problems; analyzed convergence limitations of Newton and classical continuation methods under multiple branches and turning points; adopted pseudo-arclength continuation to track folds and difficult branches; implemented topological-derivative criteria for inclusion insertion and material modification; built XPINN solvers for steady and time-dependent variable-coefficient heat conduction; consolidated weak formulations, interface terms, and finite element discretization of the linear Kapitza problem; clarified the role of PINN/XPINN methods as complementary tools for heterogeneous media, surrogate modeling, and optimization-driven simulations.

Training

Doctoral training and scientific activities

Completed training in problem-solving tools, random modeling, French A1/A2, scientific Python and JupyterLab, CIMPA Mathematics-AI school, and advanced mechanics of solids, fluids, structures and flexible multibody systems; participated in FreeFEM Days, ANR NumOpTES meeting, a winter school on control/optimization/AI, and LMAC doctoral seminars.

Funded Research Projects

2023–2026	Dynamic Multi-Fidelity Reduction Theory and Methods for Large-Scale Structural Dynamics Topology Optimization NSFC, Grant No. 12272302
	Contributed to reduced-order modeling strategies for transient structural dynamics topology optimization, including primal-dual model reduction and adaptive basis enrichment.
2021–2022	Models, Theory and Methods for Scientific Computing High-end Foreign Expert Introduction Program, Grant No. G2021183022L
	Participated in international collaboration on computational mechanics, numerical optimization, reduced-order modeling, and machine-learning-assisted simulation.
2021–2022	Large-Scale Stress-Constrained Topology Optimization Based on Reduced-Order Models Shaanxi Natural Science Basic Research Program, Grant No. 2021JM-043
	Worked on approximate reanalysis and on-the-fly reduced-order modeling for stress-constrained topology optimization.
2019–2022	Integrated Design Methods for Functionally Graded Sandwich Structures for Lightweight Thermal Protection NSFC, Grant No. 11972166
	Studied topology and material-layout design methods connecting structural mechanics, heat transfer, and gradient-material modeling.

Conference Presentations

- Invited speaker, “3D On-the-fly Reduced Order Model for Large Scale Dynamic Structure Topology Optimization,” NPU-UTC Bilateral Symposium on Data-driven Simulation, Northwestern Polytechnical University, Xi’an, 2023.
- Invited speaker, “Research on Dynamic Topology Optimization Methods Based on Reduced Models,” 13th National Conference on Random Vibration Theory and Applications and 11th National Conference on Random Dynamics, Dalian University of Technology, 2023.
- Poster presentation, “Primal-dual On-the-fly Reduced-Order Modeling for Large-Scale Transient Dynamic Topology Optimization,” ACCM-2023, 6th Australasian Congress on Computational Mechanics.

Teaching and Academic Instruction

2022–2023	Lecturer, Sino-Thai Program Xi’an International Studies University
	Delivered undergraduate instruction in Digital Economy, Statistics, Computer Design, and Digital Commerce; 256 credit hours.
2020–2023	Teaching Assistant, Joint Undergraduate Program Northwestern Polytechnical University - Queen Mary University of London
	Assisted English-taught courses in Computational Methods, Numerical Analysis, and Mathematical Modeling.
2021	Teaching Assistant, Graduate Course in Computational Methods Northwestern Polytechnical University
	Supported problem sessions, numerical exercises, and coursework feedback.
2020–2025	Academic workshop and international program assistant Selected invited lecture series and short courses
	Supported workshops on topology optimization, reduced-order modeling, machine learning in computational mechanics, high-performance computing, probability, Fourier analysis, PDEs, and quantitative finance.

Honors and Awards

- Best Paper Award, ACCM-2023, 6th Australasian Congress on Computational Mechanics.
- Outstanding Graduate, Northwestern Polytechnical University, Master's, 2024.
- First-Class and Second-Class Graduate Scholarships, Northwestern Polytechnical University, 2021–2023.
- Outstanding Graduate, Northwestern Polytechnical University, Bachelor's, 2021.
- Second Prize, National Finals of Chinese Mathematics Competitions for College Students, 2021.
- Meritorious Winner, Mathematical Contest in Modeling, USA, 2020.
- First Prize, National Mathematical Modeling Competition, Shaanxi Province, 2019.
- Third Prize, Huawei Cup Graduate Mathematical Modeling Competition, 2021.

Technical Skills

- Numerical computing and scientific programming: finite element methods, topology optimization, reduced-order modeling, FreeFEM, MATLAB, Python, and LaTeX.
- Research workflows: PDE modeling, numerical experiments, academic writing, presentation preparation, and reproducible computational studies.